

0.1 Section 6.3 - Permutations and Combinations

Let's start with permutations

0.1.1 Permutations

Suppose you have 8 different books. How many ways can the books on a shelf be arranged if...

1. The shelf fits all 8 books?

$$\underline{8} \cdot \underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 8! = \boxed{40320}$$

2. The shelf fits 5 books

$$\underline{8} \cdot \underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = \frac{8!}{3!} = \frac{8!}{(8-5)!} = \boxed{6720}$$

Here, the order matters. We call this a **permutation**

Definition 1 (Permutation)

A **Permutation** is one specific way a group of items can be arranged where order matters. In general, if we have n objects and we want to arrange k of them, then the number of ways is given by

$${}_nP_k = \frac{n!}{(n-k)!}$$

Example 1

How many ways can the president, vice president, and treasurer be assigned from a group of 4 people?

$$\underbrace{4}_P \cdot \underbrace{3}_{VP} \cdot \underbrace{2}_T = \boxed{24}$$

or, using the formula

$${}_4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = \boxed{24}$$

Now let's move on to Combinations

0.1.2 Combinations

Example 2

How many committees of 3 people can be formed from 4 people?

Here, order **DOES NOT** matter

Here, there are 4 different committees

ABC	ABD	ACD	BCD
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You can arrange the 3 people 6 different ways, given by

$${}_3P_3 = 3! = 6$$

Note

I would write everything out but I'm too lazy

Another way of getting that there are 4 committees is

$$\frac{\text{total \# ordered arrangements}}{\text{\# ways of permuting that arrangement}} = \frac{24}{6} = \frac{\frac{4!}{(4-3)!}}{3!} = \frac{4!}{3!(4-3)!}$$

This is the number of **Combinations**

Definition 2 (Combination)

A **Combinaton** is just an arrangement where order does not matter.

The number of combinations of k items taken from n items is

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

You would say "n choose k"

Think of this as "how many ways can you arrange k things in n slots"

Example 3 1. How many ways can 2 cards be dealt from a standard deck of cards?

52 cards, want 2 cards

$$\begin{aligned}
\binom{52}{2} &= \frac{52!}{2!52-2!} \\
&= \frac{52!}{2!50!} \\
&= \frac{52 \cdot 51}{2} \\
&= \boxed{1326}
\end{aligned}$$

2. A coin is flipped 10 times. How many ways can exactly 4 of the flips be heads?

$$\begin{aligned}
\binom{10}{4} &= \frac{10!}{4!6!} \\
&= \boxed{210}
\end{aligned}$$

notice how this also works for 6 flips to be tails!

3. Ten distinct points are labeled on a circle. How many quadrilaterals can be drawn using these points?

$$\binom{10}{4} = \frac{10!}{4!6!} = \boxed{210}$$

There's a LOT more examples but I'm too lazy to write them all out. Just go back to the notes and write them out later

Let's prove that $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

Proof

$$\begin{aligned}
\binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!(n-(k-1))!} + \frac{n!}{k!(n-k)!} \\
&=
\end{aligned}$$

uhhh do later (it's on the very back of 6.3)